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Una serie di accorgimenti sulla sintassi e semantica delle policy di bgp

1) Non specificare più di una prefix list per lo stesso tipo di traffico (in o out) per lo stesso neighbor, lo stesso vale per le route map. l'effetto è che la seconda sovrascrive la prima. Questo non è un problema , perche all'interno dello stesso prefix list e route map possiamo mettere quanti statement vogliamo.

frr configuration file

```
router bgp 100
!
neighbor 11.0.0.2 remote-as 20
neighbor 11.0.0.2 description Router as20r2 (primary)
!
network 100.1.0.0/16
!
neighbor 11.0.0.2 prefix-list mineOutOnly out
neighbor 11.0.0.2 prefix-list acceptDefault in
```

Anche questo perfettamente legale.

frr configuration file

```
router bgp 100
!
neighbor 11.0.0.2 remote-as 20
neighbor 11.0.0.2 description Router as20r1
!
network 100.1.0.0/16
!
neighbor 11.0.0.2 prefix-list mineOutOnly out
…
…
ip prefix-list mineOutOnly permit 100.1.0.0/16
…
```

prefix-list attachment to peering

prefix-list definition

Per vedere se torna quello che abbiamo dichiarato, fare su vtysh uno show running-config.

Facciamo per esempio un errore nella definizione della prefix list:

```
!
neighbor 11.0.0.2 description Router as20r2 (primary)
neighbor 11.0.0.2 prefix-list mineOutOnly out
!
neighbor 11.0.0.2 remote-as 20
```

Frr non ti dice nulla! Non te ne accorgi, ma per accorgersene puoi andare nella console e guardi i comandi dalla console, e scopri che i comandi che hai inserito non ci sono!

```
!
router bgp 100
 neighbor 11.0.0.2 remote-as 20
 neighbor 11.0.0.2 description Router as20r1
 !
 address-family ipv4 unicast
  network 200.2.0.0/16
 exit-address-family
!
ip prefix-list mineOutOnly seq 5 permit 200.2.0.0/16
ip prefix-list acceptDefault seq 5 permit 0.0.0.0/0
!
line vty
!
end
```

prefix-lists missing

Sintassi prefix list:

- a **prefix-list** can be composed of different statements
- **prefix-list** statements can be ordered according to a sequence number
  - examples:

frr configuration file
```
ip prefix-list myPfxList seq 5 permit 10.0.0.0/8
```

frr configuration file
```
ip prefix-list myPfxList seq 5 permit 10.0.0.0/8
ip prefix-list myPfxList seq 10 permit 20.0.0.0/8
```

Se non inserisci un numero di sequenza, i seq vengono inferiti automaticamente da frr.

- for example

frr configuration file
```
ip prefix-list myPfxList permit 10.0.0.0/8
ip prefix-list myPfxList permit 20.0.0.0/8
```

- is equivalent to

frr configuration file
```
ip prefix-list myPfxList seq 5 permit 10.0.0.0/8
ip prefix-list myPfxList seq 10 permit 20.0.0.0/8
```


prefix-list matchings

- the prefix of a **prefix-list** statement must be matched exactly
- for example

frr configuration file
```
ip prefix-list myPref permit 10.0.0.0/8
```

matches the announcements of:
- 10.0.0.0/8 exact match

does not match the announcements of:
- 10.0.0.0/24
- 10.10.10.0/24

more specific: not matched

- 10.0.0.0/7 less specific: not matched

Ogni prefix list termina con un deny all, implicitamente.

- referencing to an undefined **`prefix-list`** in a neighbor statement is equivalent to denying anything
  - for example, this command…

Questo può capitare se facciamo errori di battitura!

Route map:

frr configuration file

```
router bgp 100
!
neighbor 11.0.0.2 remote-as 20
neighbor 11.0.0.2 description Router as20r2
!
network 100.1.0.0/16
!
neighbor 11.0.0.2 route-map myRouteMapOut out
neighbor 11.0.0.2 route-map myRouteMapIn in
```

route-map attachment to peering

- a **`route-map`** definition may consist of multiple statements
- each statement has the following syntax

```
route-map <route-map-name> permit/deny <priority>
  match <condition>
  match <condition>
  …
  match <condition>
  set <action>
  set <action>
  …
  set <action>
```

zero or more (match)

zero or more (set)

frr configuration file

```
router bgp 100
!
neighbor 222.2.2.2 remote-as 200
!
```

route-map attachment to peering

```
neighbor 222.2.2.2 route-map myRouteMap in
!
route-map myRouteMap permit 10
  match ip address myAccessList
  set metric 5
  set local-preference 25
!
route-map myRouteMap permit 20
  set metric 2
!
access-list myAccessList permit 193.204.0.0/16
```

route-map definition

Se la match condition è assente, tutto cio che viene processato, viene considerato come candidato.

frr configuration file
```
route-map myRouteMap permit 10
set local-preference 200
```

sets local preference on all the announcements

route-map without set conditions

- if set conditions are absent announcements are unchanged
  - for example

frr configuration file
```
route-map myRouteMap permit 10
match as-path _10_
```

allows only annuncements containing as 10 in the as-path

  - another example

frr configuration file
```
route-map permissiveRouteMap permit 10
```

allows any announcement

- typical set conditions are:
  - **set as-path prepend**
    - add numbers to the as-path
    - example: `set as-path prepend 10 10 10`
  - **set local preference**
    - changes the local pref of incoming announcements
    - example: `set local-preference 200`
  - **set metric**
    - changes the multi-exit-discriminator attribute
    - example: `set metric 10`

- **set community**
  - add a label to the outgoing announcements
  - example: **set community 200:11111**

Stessa tipologia di matching è fra loro in or, diverse tipologie di matching sono in and

frr configuration file

```
route-map myRouteMap permit 10
match ip address accessList1    } OR  } AND
match ip address accessList2    }     }
match as-path 10 20 30                }
set local-preference 200
```

frr configuration file

```
route-map myRouteMap deny 10
set local-preference 200
```

matches any announcement and filters it out (without setting any local preference)

route-maps and acls

- access-control-lists (acl) are used with the match ip address conditions

frr configuration file

```
route-map myRouteMap permit 10
  match ip address myAccessList
  set local-preference 25

access-list myAccessList permit 10.0.0.0/16
```

- differently from prefix-lists, acls match also more-specific prefixes
  - in the example **10.0.4.0/24** would also match

- in order to only match **10.0.0.0/16** you have to add **exact-match** to the access-list definition

frr configuration file

```
route-map myRouteMap permit 10
  match ip address myAccessList
  set local-preference 25

access-list myAccessList permit 10.0.0.0/16 exact-
```